






SWAYAM NPTEL
ONLINE CERTIFICATION COURSE ON
Statistical and Machine Learning Methods: Basics and Applications

MODULE 01 : Role of Statistics, Machine Learning and Data science
LECTURE 01 : Introduction to Statistical and Machine Learning

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CONCEPTS COVERED

Basic Introduction: Complementary Role of Statistical and Machine Learning

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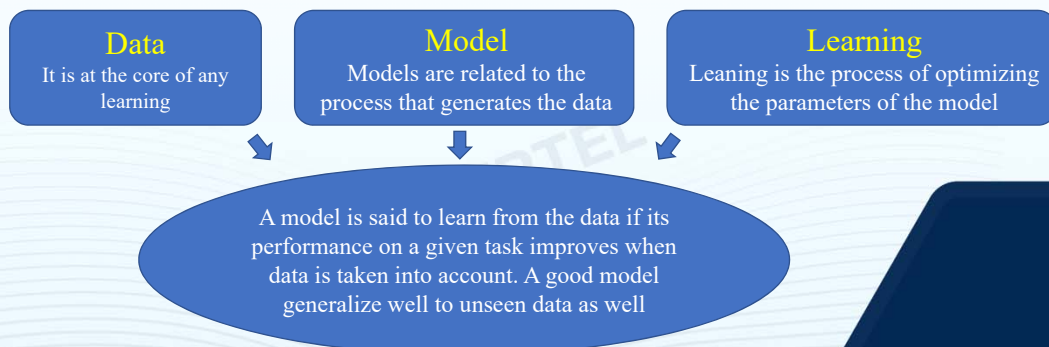
Statistical and Machine Learning

- **Statistical and/or Machine Learning** is all about methods or algorithms that extracts valuable information from data.
- Is it **automatic**? No.
 - General purpose of developing statistical and machine learning models is to be applicable to data sets apart from the data sets that was used to trained the model.



Statistical and Machine Learning: Data, Model and Learning

- There are three interlinked aspects:



Statistical and Machine Learning: Data, Model and Learning

- **Data** → Most precious in this domain. We generally treat the data as a *Vector*.
- **Model** → Model is supposed to describe/represent the underlying *process* to capture. It is used for generating data preserving the information available in the available dataset.
- **Learning** → It is all about estimating the parameter (static or dynamic) using the available data with respect to a *utility function* that evaluates how well the model capture the *underlying process* so that it fits well to *training* as well as *unseen data*.



Statistical and Machine Learning: Overall Mathematical Background

- Suppose, we have a quantitative response Y and p different predictors, $X_1, X_2, \dots, X_p$.

$$Y = f(X) + \epsilon$$

here f is a fixed but unknown function. It represents the systematic information that X provides about Y .

- ϵ is a random error term, which is independent of X and has mean zero.
- The function f helps in prediction or simulation or inference.

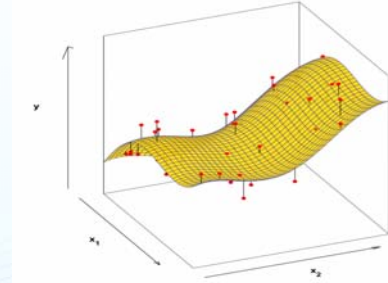


Statistical and Machine Learning: Overall Mathematical Background

- Prediction:

$$\hat{Y} = \hat{f}(X) + \epsilon$$

$$\begin{aligned} E(Y - \hat{Y})^2 &= E(f(X) + \epsilon - \hat{f}(X))^2 \\ &= \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{Reducible error}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible error}} \end{aligned}$$



- Our focus is to estimate the f with the aim of minimizing the **reducible error**.



Statistical and Machine Learning

Some questions that we are generally interested to know:

- What are the predictors and corresponding response? Is there any relationship?
- Is the relationship linear? What are the underlying process(es)?
- What is data? Is the data sufficient for the inference?

⋮

- **A Point to Remember:** The **meaningful information** include the information as simple as **basic statistics** or as complex as **inference** based on the partial information or **decision** on a **hypothesis**.



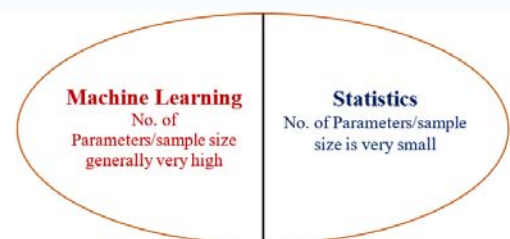
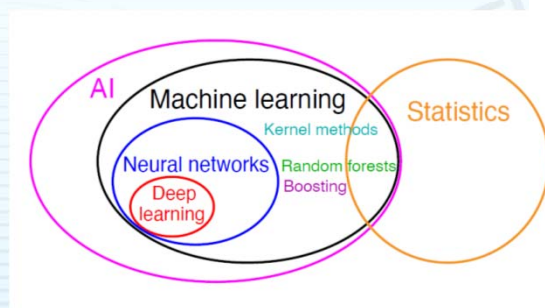
Statistical and Machine Learning

- Major difference between the statistics and ML cultures lies in their treatment of predictor variables.
 - **Predictor selection**, that is, choosing only the relevant predictor variables from a pool of predictors, is commonly practiced in statistics but not often in Machine Learning.
 - Both Statistical and Machine Learning generally **does not consider throwing away information** a good practice. This is also not the practice in statistical methods as well but often used in some simpler applications.
- Selection of predictors based on correlation/association with the response variable, may lead to better performance of Machine Learning models. However, physical or conceptual justification is always recommended while selecting the predictors.



Statistical and Machine Learning

Statistical and Machine Learning models in the perspectives of parameter/sample size:



CONCLUSION

- In summary, the main tradeoff between statistics and Machine Learning is **interpretability** versus **accuracy**.
 - With relatively few parameters and few predictors, statistical models are **much more interpretable** as compared to Machine Learning models.
- For example, the parameters in a **linear regression** model give useful information on how each predictor variable influences the response variable.



CONCLUSION

- Machine Learning methods, such as artificial neural networks and random forests, are run as an ensemble of models initialized with different random numbers, leading to a huge number of parameters that are **uninterpretable** in practical problems.
- However, as datasets become **increasingly larger and more complex**, interpretability becomes harder and harder to achieve even with statistical models, while the advantage in prediction accuracy attained by Machine Learning models makes them increasingly attractive.










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Statistical and Machine Learning Methods: Basics and Applications

MODULE 01 : Role of Statistics, Machine Learning and Data science
LECTURE 02 : Statistical and ML with Data Science

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CONCEPTS COVERED

Role of Statistics, Machine Learning and Data science

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Statistical and Machine Learning

Some Points to Remember:

- **Data science** deals with the studies related to developing models and tool using both statistical and machine learning algorithms that helps us to extract meaningful information from a (generally, huge) data set.
- The **meaningful information** include the information as simple as **basic statistics** or as complex as **inference** based on the partial information or **decision** on a **hypothesis**.



Overview

- The term **Data Science** was coined in 1990s by the statisticians. Now-a-days, it is considered as the “connective tissue” between raw information and actionable intelligence.
- At its core, it is an interdisciplinary field that synthesizes principles from **mathematics**, **statistics**, **computer science**, and **domain expertise** to uncover hidden patterns and insights within complex datasets.
- Data science helps in extracting meaningful information from such a **huge data sets**. However, developing the information/data for decision or design is challenging.
- It needs **specialized skills** to tap from the available data using data analysis, statistical and machine learning algorithms.



Overview

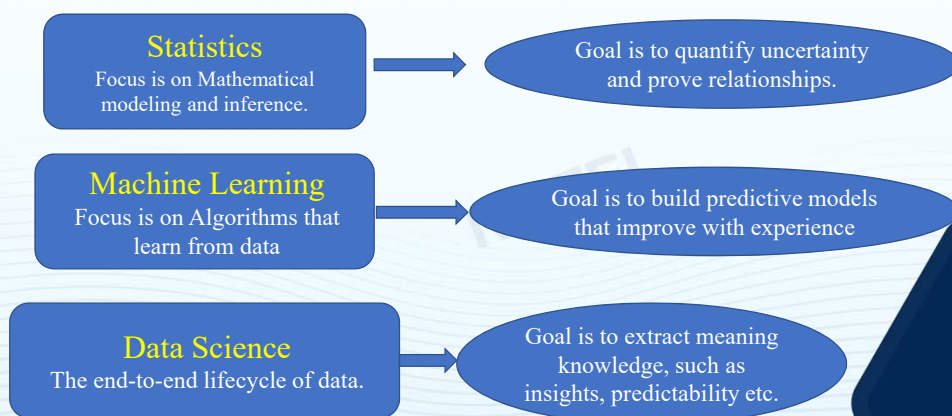
There are **three** foundational disciplines:

- **Statistics and Mathematics:** This is the analytical engine. Techniques like linear algebra, calculus, and probability theory provide the framework for understanding data distributions, testing hypotheses, and quantifying uncertainty.
- **Computer Science and IT:** These provide the tools for scale. This includes data engineering (how we store and move data), algorithm design, and the high-performance computing required to process “Big Data”.
- **Domain Expertise:** This is often the most overlooked pillar. Without a deep understanding of the specific industry (e.g., finance, healthcare, meteorology, or hydroclimatology), a data scientist cannot ask the right questions or interpret the results meaningfully.



Statistical or Machine Learning

- The focus and role of each discipline:



CONCLUSION

- In summary, **Statistics** and **Machine Learning** are essential components of the data science toolkit.
- **Statistics** provides the theoretical rigor, while **Machine Learning** provides the predictive power.
- In the past, data was scarce and expensive to store. Now, there is huge data to explore.
- However, in some cases/situations, there is NO data and the underlying processes are unknown.
- Recent technological advancement has led to a tremendous growth in observed and simulated data in various domains, including engineering, finance, earth sciences etc.



CONCLUSION

- There are many allied engineering fields. We will mostly focus in the field of hydrology, climatology and earth science, in general. However, the discussion will be general to understand the basics and to apply to any other fields of applications.










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Statistical and Machine Learning Methods: Basics and
Applications
MODULE 01: Role of Statistics, Machine Learning
and Data science
LECTURE W1L3: Role of Statistical and Machine Learning
Methods in Interdisciplinary studies
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CONCEPTS COVERED

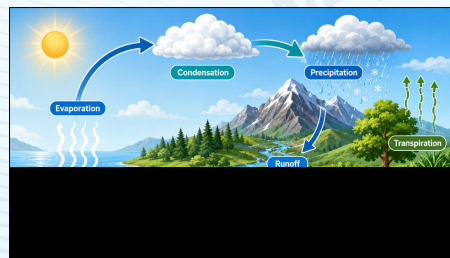
**Discussion on Statistical and Machine Learning
 Methods in Interdisciplinary Studies**

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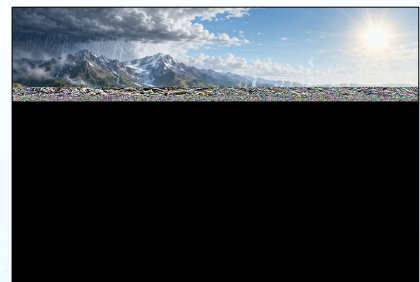
The Area We Take as an Example: **Hydroclimatology**

- **Hydrology** is the science that involves studies related to the occurrence and movement of water (in any phase of solid, liquid or vapor) in the combined system of surface, subsurface and atmosphere. Hydrologic cycle, also known as water cycle, is the basis of the hydrologic science.
- It offers a platform to manage the available water in the context of water use, water control and water pollution.



The Area We Take as an Example: **Hydroclimatology**

- **Climatology** is the field of study related to exchange of mass, momentum and energy between land/ocean surface and atmosphere.
- Vertical and horizontal fluxes of these quantities drive the interaction between earth surface (both land and ocean) and atmosphere. These fluxes also control the atmospheric circulation at different scales.
- **Atmospheric component of hydrologic cycle** is coupled with climatic phenomena, and thus, any change or variability may affect each other through different feedback systems.



The Area We Take as an Example: **Hydroclimatology**

- *Hydroclimatology* is an interdisciplinary area of study that deals with the interaction between hydrology and climatology to identify the influence of the climatic system on different hydrologic processes, which are the parts of hydrologic cycle.
- For example, hydrologic variables, such as rainfall, soil moisture, and streamflow etc. are significantly influenced by various global or local scale atmospheric circulations.



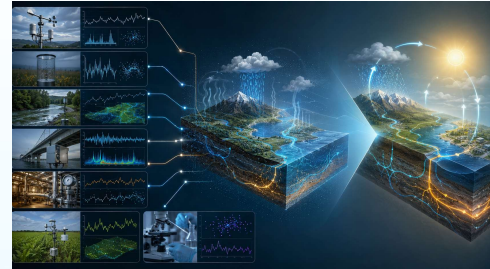
Statistics and ML Methods in Interdisciplinary Studies

- Statistical and machine learning methods are widely employed to gain valuable insights across various fields.
- Statistics and machine learning provide the methods needed to combine heterogeneous datasets and quantify uncertainty.
- In scientific and applied engineering fields, the core physical mechanisms are often analyzed depending upon required detail and scale.
- In any field of studies, the data from other fields may also be beneficially used to reveal insights of the underlying physical process.



Statistics and ML Methods in Interdisciplinary Studies

- Statistical or machine learning approaches help to capture the interrelationships or the underlying processes.
- Therefore, the data from various sources needs to be ensembled through a model.
- The selected model may be a statistical model or a machine learning approach.
- Therefore, the insights from these approaches are indispensable in the context of interdisciplinary studies.



Statistics and ML Methods in Interdisciplinary Studies

Random process and uncertainty:

- The associated uncertainty should be treated through statistical methods involving the associated random variable.
- For example, uncertainty is associated with the variables such as: annual maximum flow, annual maximum rainfall etc.,

Role of statistical methods:

- Uncertainty quantification
- Frequency analysis
- Forecasting along with uncertainty quantification



Combined role of Statistical and Machine Learning Methods in Interdisciplinary Studies

- **Data Driven Models** explore the mutual relationships between input variables and target variable rather than encoding **physical laws** explicitly by inferring the patterns from the observed data.
- Reliable prediction depends on the best set of predictor selection that depends on **physical laws**.
- Both **domain knowledge** and **combined approaches** are needed to capture the complex underlying association.
- It helps in extracting the **essential information/insights** from the data, and **mutual relationship** between the input and target variables.



Curse of Dimensionality in Interdisciplinary Studies

- The **curse of dimensionality** refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces (spaces with many features or variables) that do not occur in low-dimensional settings.
- A large predictor set makes model calibration difficult and may lead to **overfitting**. Multiple inputs may convey **overlapping information**, if two predictors are highly correlated.
- Including the second input may add little value but increase **complexity** and **redundancy**.
- A combination of statistical and Machine Learning methods helps in dimensionality reduction before training models. For example, **Feature Engineering and Feature Selection**, **Principal Component Analysis**, etc.










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Statistical and Machine Learning Methods: Basics and Applications

MODULE 01 : Role of Statistics, Machine Learning and Data science

LECTURE 04 : Data Type and Related Issues

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CONCEPTS COVERED

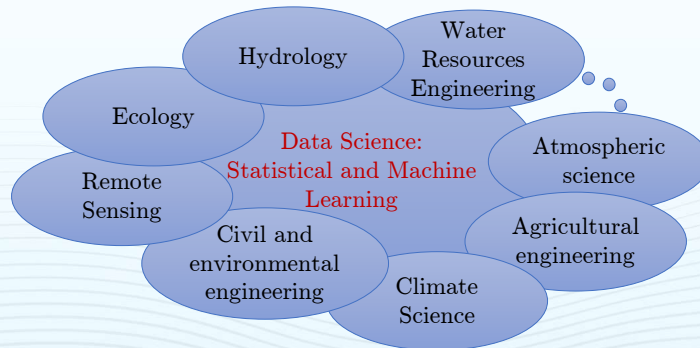
Data Type and Related Issues

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Statistical and Machine Learning

- **Scientific data and data science** form a crucial interaction for any analysis to reveal insights.

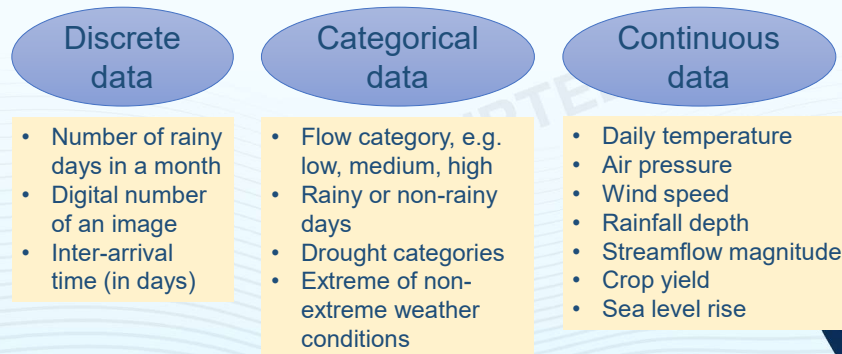


Data from different scientific/engineering domain have their own characteristics.



Types of data

- Statistical methods have long been popular in these domains of science and engineering
- We need treat two types of data differently:



A Few Points to Remember

- The Intersection of Disciplines
- Continuous vs. Discrete Data
- Bounded vs. Unbounded Domains
- Regression Dominance
- The Extrapolation Challenge
- The Impact of Data Averaging
- A Shift Toward Extremes
- Uneven ML Adoption Across Disciplines
- The Challenge of Data Scarcity
- The Convergence of Physics and ML



Concluding Remarks

- Sometime interdisciplinary data science requires distinct approaches compared to traditional ML, as it primarily deals with unbounded, continuous variables (like wind speed and temperature) rather than bounded, discrete categories.
- There is a risk of extrapolation. The unbounded nature of data makes predictive modeling highly susceptible to extreme outliers, leading to severe extrapolation errors when models encounter unprecedented conditions.
- Non-linear ML models excel at short-term forecasting but lose effectiveness when applied to long-term data, as statistical averaging over time weakens non-linear relationships.



Concluding Remarks

- In some fields, simple time averages may be problematic. For example, dealing with climate extremes, like droughts or intense rainfall, which are affected by global climate change.
- While adoption rates historically varied based on data availability and the success of existing tools, traditional physics-based models and advanced ML techniques are now rapidly converging to create more powerful, interpretable models.








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Statistical and Machine Learning Methods: Basics and Applications

MODULE 01 : Role of Statistics, Machine Learning and Data science
LECTURE W1L5 : Uncertainty and Data Visualization

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CONCEPTS COVERED

Uncertainty and Data Visualization

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Concept of Uncertainty

- **Uncertainty** refers to a situation where information about some natural or real-world system is incomplete, imprecise, or imperfect.
- Statistical or Machine Learning models are based on **assumptions and simplifications** of real-world systems, for which available information is often incomplete, imprecise, or uncertain.
- Thus, concept of uncertainty is important to understand as natural processes exhibit variability that cannot be predicted exactly.
- Decisions must be made even when **future outcomes** are **not known with certainty**.
- Therefore, uncertainty is an **inherent part** of engineering analysis, planning, and design.



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Sources of Uncertainty

1. Inherent randomness:

- Most of the real-world processes, if not all, vary naturally over time and space.
- Influenced by physical, chemical, biological, and social factors, etc.
- Cannot be completely removed.
 - Examples: Spatiotemporal variability in rainfall, flood etc.
- This uncertainty is unavoidable.

2. Model uncertainty:

- Models simplify real-world processes.
- Assumptions are required to represent complex systems.



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Sources of Uncertainty

- Simplifications may not fully capture actual behavior.
- It can be reduced to some extent by improving the physical representation of processes.

3. Parameter Uncertainty:

- Model parameters are estimated during calibration.
- This uncertainty is caused due to:
 - Inaccurate parameter estimation
 - Changes in system conditions
- It can be reduced to some extent by improving physical representation of processes and using more realistic models.



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Sources of Uncertainty

4. Data Uncertainty:

- Due to data handling errors, measurement errors, inconsistent data and non-homogeneity in data.
- It can be avoided by improving the data quantity and quality through improved data collection and data handling.

5. Operational Uncertainty:

- Arises from human errors during the implementation or execution of a design.
- Includes variability introduced during manufacturing, construction, and maintenance.
- Can be minimized through good workmanship, and effective quality control practices.



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Types of Uncertainty

Aleatory Uncertainty:

- Originates from inherent natural variability and randomness.
- Cannot be eliminated through additional information.
- It is considered as irreducible.

Epistemic Uncertainty:

- This type of uncertainty results from incomplete knowledge, data limitations, and measurement inaccuracies.
- Can be reduced through improved observations, data collection, and understanding



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Analysis of Uncertainty

Task involved:

- Quantification the associated uncertainty.
- Assessment of the confidence level that can be placed in model outputs.
- Determination of the contributions of different input variables and model parameters to the overall output uncertainty.
- Identification of the most sensitive inputs and parameters that significantly influence model performance.

Components of uncertainty analysis

Qualitative

Quantitative

Communication of Uncertainty



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Components of Uncertainty

Qualitative Uncertainty Analysis:

Identifies and classifies the different sources of uncertainty present in the system.

Quantitative Uncertainty Analysis:

Measures the magnitude of uncertainty and evaluates its impact on model outputs using statistical and machine learning models.

Uncertainty Propagation:

Explains how uncertainties in model inputs and parameters propagate through the model and influence the final outputs.



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Measures of Uncertainty

➤ Variance:

- One of the most common measures of uncertainty is variance.
- Variance describes how far individual observations are spread around their mean value.
- Small variance indicates observations are clustered closely around the mean and large variance indicates observations are widely dispersed.

➤ Confidence Interval:

- A confidence interval expresses uncertainty by providing a range within which the true value of a variable is expected to lie with a specified level of confidence.
- It will be discussed in detail in the subsequent lectures.



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Measures of Uncertainty

➤ Quartiles and Inter-Quartile range:

- Uncertainty can also be described non-parametrically using quartiles, which divide an ordered dataset into four equal parts.
 - First Quartile (Q1):** 25% of observations lie below this value.
 - Second Quartile (Q2):** 50% of observations lie below this value and is equivalent to the median.
 - Third Quartile (Q3):** 75% of observations lie below this value.
- The difference between Q3 and Q1 is called the **Inter-Quartile Range (IQR)** and represents the spread of the central 50% of the data.



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Graphical Representation of Data

Why to visualize data?

- Data visualization** is a fundamental step in statistical analysis and machine learning.
- Graphs provide a quick understanding of **trends, patterns, variability, and anomalies in data**.
- Important characteristics of a dataset are **easier to identify visually** than numerical summaries alone.
- Different datasets may produce **similar statistical measures** but exhibit completely **different underlying patterns**.



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Why visualize Data?

➤ Data visualization helps to identify:

- Outliers
- Missing values
- Skewness
- Trends and Seasonality
- Even to some extent, the relationship between variables

Examples:

- Rainfall variability over time
- Streamflow trends and extremes
- Relationships between climate variables and hydrological responses



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Representation of Data

- Most time series are inherently continuous, but for analysis and storage purposes, they are represented at discrete time intervals.
- For example, variables such as temperature, streamflow, and rainfall vary continuously with time, although observations are recorded at specific intervals.
- A continuous time series (let's denote it as, $f(t)$) can generally be represented in discrete form using two approaches.
 - Sample data representation
 - Pulse data representation



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Representation of Hydroclimatic Data

Sample Data Representation

- The value recorded for a time interval corresponds to the instantaneous value of the variable at a specific instance of time.
- Commonly used when measurements are taken at an instance with a regular interval of time.
 - Examples: Streamflow measurement, Air temperature records, Wind speed and Water level measurements, etc.



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Representation of Hydroclimatic Data

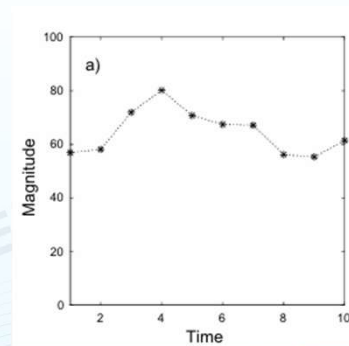
Sample Data Representation

- In this representation, value of the function for i^{th} time interval, $X(i)$, is given by instantaneous value of $f(t)$ at the time $i\Delta t$.

$$X(t = i) = f(i\Delta t)$$

- Dimensions of the sample data is:

$$L^3T^{-1} \text{ or } LT^{-1}$$

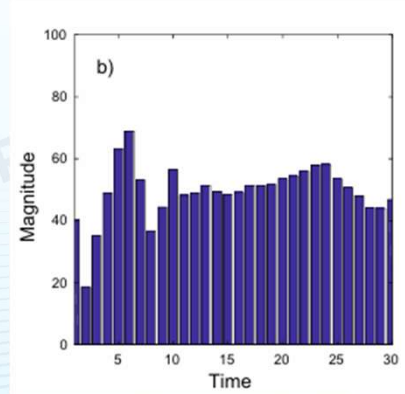


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Representation of Hydroclimatic Data

Pulse Data Representation

- The recorded value represents the total amount accumulated over a specified time interval.
- Represents the cumulative effect during that specified time interval.
- The value depends on the entire interval rather than a single instant.
- Frequently used for variables that accumulate over time.
- Dimension of the pulse data is L^3 or L .



Representation of Hydroclimatic Data

Pulse Data Representation

- In this representation, value of the function for i^{th} time interval, $X(i)$, is given by,

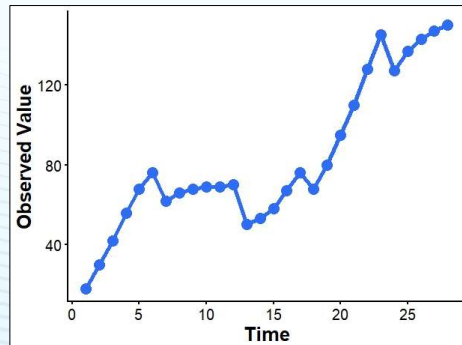
$$X(t = i) = \int_{(i-1)\Delta t}^{i\Delta t} f(t)dt$$

- Most common example of pulse data representation is **precipitation** that is recorded as a series of accumulated depths.
- Sometimes, pulse data can also be represented as average rate over the interval Δt as follows,

$$X(t = i) = \frac{1}{\Delta t} \int_{(i-1)\Delta t}^{i\Delta t} f(t)dt$$

Time Series Data

- Time series data is a sequence of observations recorded in chronological order at regular or irregular time intervals.
- It helps us understand how a variable changes over time and is widely used in many field of studies, including hydroclimatology, finance, engineering, health care studies, etc.

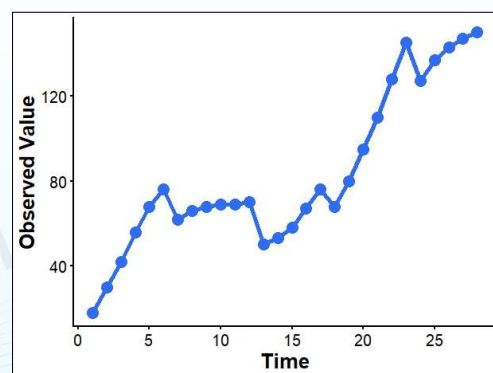


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Time Series Data

Key characteristics of time series data include:

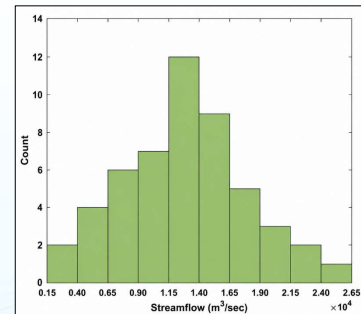
- **Trend:** Long-term increase or decrease in the data.
- **Seasonality:** Repeating pattern within a fixed time period.
- **Cyclic variation:** Long-term repeated fluctuations without a fixed period.
- **Random variation/noise:** Irregular fluctuations caused by uncertain or unexplained factors.



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Histogram

- Graphical representation used to **display the distribution** of numerical data.
- Data values are grouped into a series of **intervals** (bins) of equal width.
- Height of each bar represents **frequency**.
- Helps in **visualizing** the descriptive statistics of data (e.g., **mean, variance, skewness, etc.**).
- Simple to interpret and provides a quick overview of data distribution.



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Histogram

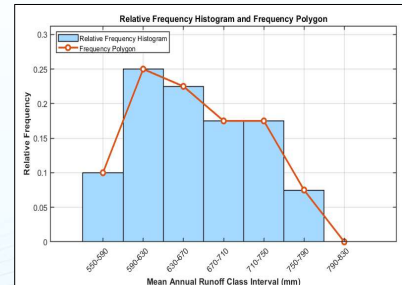
- Effective for representing **discrete and categorical count of data**.
- Some of the limitations are:
 - Appearance depends strongly on the number and width of bins selected.
 - Different bin sizes can produce different visual representations of the same dataset.
 - Exact data values cannot be identified from the plot.
 - Important distributional features may be hidden when continuous data are grouped into wider intervals.



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Relative Frequency from Histogram

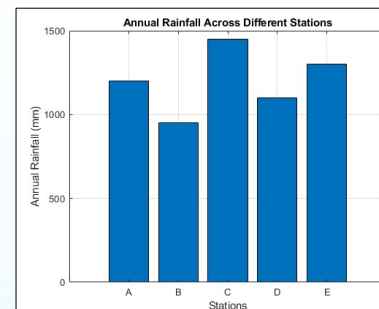
- Relative frequency is the graphical representation of **relative frequencies** across the class intervals.
- Helps in determining the **shape of the distribution** of the variable. Sometimes, called as frequency polygon.
- Useful for comparing multiple datasets on the same graph and allow comparison between datasets of different sizes.
- Widely used in many fields of studies.
- In hydroclimatology, it is frequently used for rainfall, streamflow, temperature, and drought analyses.



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Bar Plot

- Bar plots are used to compare values across different categories or groups.
- Each category is represented by a vertical or horizontal rectangular bar of equal width.
- Height (length, in case of horizontal bar plots) of the bars vary depending on the size of data values and represents the **magnitude** of the values of the variable.
- Bars are separated by gaps, indicating that the categories are distinct and independent.
- Width of bars does not convey anything (as compared with histograms, where it matters)



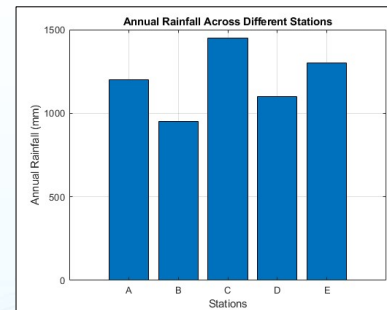
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Bar Plot

- Used for categorical or discrete data.
- The categories can be arranged in any order.

Examples:

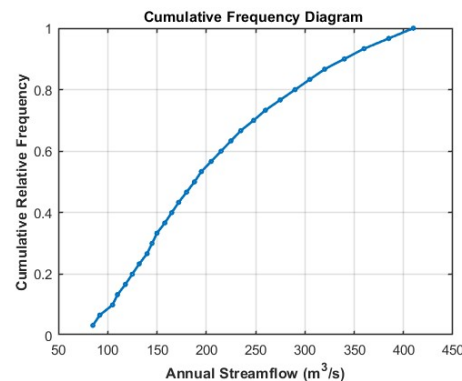
- Mean annual rainfall at different stations
- Streamflow from different basins
- Water demand across different sectors
- **Note:** Bar plots are used to compare values across categories or groups, whereas histograms are used to visualize the frequency distribution of a continuous variable.



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Quantile Plots

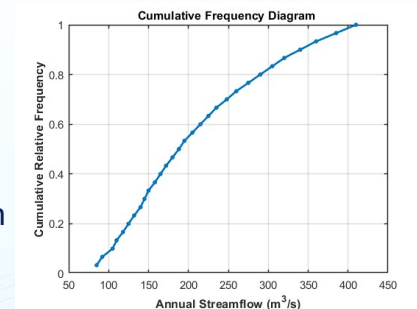
- Graphical representation of the **cumulative distribution** of a dataset.
- Also known as Empirical Cumulative Distribution Function (ECDF).
- Represents the probability that a random variable is less than or equal to a specified value.
- Provides the quantiles (or percentiles) of a dataset in a graphical form.



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Why use Quantile Plots?

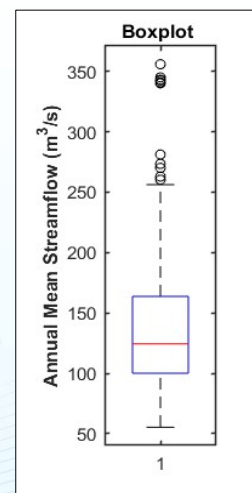
- No arbitrary bin selection (unlike histograms).
- All observations in the data are displayed.
- Easy estimation of **percentiles** and **quantiles**.
- Each data point occupies a unique location on the graph without overlapping other points.
- Useful for comparing multiple datasets.
- Forms the basis of: **Q-Q Plots**, **Probability Plots**.
- Useful for identifying **clusters** of observations and **extreme** values.



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Box Plot

- A **boxplot** is a concise graphical display used to **summarize the distribution** of a dataset.
- It provides information on:
 - Center of the data (median)
 - Variation or spread (interquartile range)
 - Skewness (relative size of box halves)
 - Presence and magnitude of unusual values (outliers)
- **Boxplots** are particularly useful for comparing these characteristics among several datasets.



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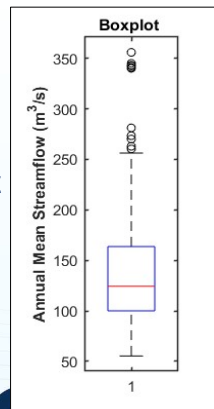
Box Plot

Components:

- Box representing the middle 50% of the data, with the boundaries at the 25th and 75th percentiles
- Median line within the box
- Whiskers extending beyond the box (extend up to 1.5 times the IQR)
- Outside values shown as individual symbols

Limitation:

- Provide limited information about the characteristics of the middle portion of the data.
- Features, such as bimodality, are not be evident.



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Thank You!

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